Standalone Step 3: information, actions and complete gradients

19 October 2026 — selected H1/T1, S-E-P, R-E, O-R and W-F law

Scope. The user makes SUMO optional. The defined conditional model is the present verification target. This note changes no equation, action, observation, queue guard, renewal mass or completion boundary. It derives new analytical comparison and sensitivity references; it does not run a traffic trajectory, AD, numerical differences or an optimizer. Physical admission and external usefulness are unestablished. Full equations are in the 18-page consolidated model.

1 What information is missing, and what closes prediction?

Let $v \in [v_-, v_+]$, $L = L_p + b$, and let the initial homogeneous footprint be $[0, \ell]$. Densities are $e_0, a_0 = b_0$, total density ρ, mass $M = \ell\rho$, and $r = e_0 + a_0$. Physical parameters are common and known; the initial terminal and recovery histories are empty. With $K = wk_j$,

$$k_c(v) = \frac{K}{w+v}, \quad C(v) = \frac{Kv}{w+v}, \quad U(g) = \frac{wg}{K-g}, \quad e_z = -h^{-1}\min\{e, (k_c(v) - r + e)/\gamma\}. \tag{1}$$

The admitted domain includes $v_+ r < R \le C(v_-)$, finite storage $M < k_j b$, the original lane-density/far-lane bounds, and $c_0 = 1 + \delta \ge 1$, $\beta \ge 0$, $c_0 - \beta(1 + v_+/w) > 0$. The fixed coefficient is $\chi = \beta/[C(v_+)v_+]$; it is not differentiated with advice.

The earlier H1 loading construction remains valid because all predecision traffic stays upstream of the cutoff; the amended terminal is empty. For common past speed u, $T = \ell/u$, prescribe on $[-T, 0]$

$$e_i^{\text{in}}(t) = e_{0,i}e^{-ut/h}, \quad a_i^{\text{in}}(t) = r_i - e_i^{\text{in}}(t), \quad b_i^{\text{in}} = b_{0,i}. \tag{2}$$

Under the previously checked donor/loading bounds, both preparations have the same occupied front $u(t+T)$ and identical lane-blind fields

$$k_\Sigma = \rho \mathbf{1}_{0<x<u(t+T)}, \quad q_\Sigma = uk_\Sigma, \quad N_\Sigma = \rho[u(t+T) - x]_+. \tag{3}$$

Thus even their full aggregate density, speed, longitudinal-flow and cumulative-count histories cannot identify the actual initial ending mass. Lane or lateral-event observations are extra channels, not part of this collision. A lawful aggregate inference gets every admitted channel.

Given E_0, however,

$$e_0 = E_0/\ell, \qquad a_0 = b_0 = (M - E_0)/(2\ell). \tag{4}$$

The conditional unique forward law then predicts all preparation, arrivals, queues, speed marks, recovery history and departures. Aggregate-plus-E_0 (A+E) and explicit lane state (G) therefore give identical $J(v)$ for every admitted advice, and identical derivatives wherever defined. Recovery history is predicted state, not an extra initial observation. This does not prove E_0 is the only possible sufficient encoding.

2 A retained reference family and an exact negative interval

Reuse the earlier analytical H1/T1 pair, rather than S13's replaced clock:

$$\begin{gathered} v_- = u, \quad v_+ = 2u, \quad w = 2u, \quad k_j = 3\kappa, \quad \gamma = 2, \\ R = 39\kappa u/20, \quad R_B = \kappa u, \quad b = \ell, \quad L_p = \ell + h\log 2, \quad 0 < \ell/h < \log(4/3), \\ \mathcal{L} : (e_0, a_0, b_0) = (3\kappa/5, 3\kappa/10, 3\kappa/10), \\ \mathcal{H} : (e_0, a_0, b_0) = (7\kappa/10, \kappa/4, \kappa/4). \end{gathered} \tag{5}$$

Both have $\rho = 6\kappa/5$ and the identical history (3). All original domain bounds hold; in particular $C(u) = 2\kappa u$, $M/(k_j b) = 2/5$ and $R - v_+ \max_i r_i = \kappa u/20 > 0$.

Write $E_f(v) = e(L_p - \ell; v)$. The earlier explicit preparation calculation gives

$$E_{f,\mathcal{L}} = 3\kappa/10, \quad E_{f,\mathcal{H}}(v_+) = 125\kappa/352, \quad E'_{f,\mathcal{H}}(v_+) = 225\kappa/(7744u) > 0. \tag{6}$$

H1 comparison makes front work $W_f(v) = v[r + \delta E_f(v)]$ increasing with advice. At v_+,

$$\frac{W_{f,\mathcal{L}}}{\kappa u} = \frac{9}{5} + \frac{3}{5}\delta, \qquad \frac{W_{f,\mathcal{H}}}{\kappa u} = \frac{19}{10} + \frac{125}{176}\delta. \tag{7}$$

Consequently, for the nonzero interval

$$0 \le \delta \le 44/625, \qquad J_{\mathcal{L}}(v) = J_{\mathcal{H}}(v) = \frac{M(L - \ell/2)}{v}. \tag{8}$$

Both streams stay free for all advice. Equal free marks make mismatch zero and free E creates no recovery; equality at the threshold remains free under the selected guard. This result permits positive β and finite positive recovery acceleration. Hidden E mass exists but is unnecessary for the best advice in this interval. The larger predecessor small-work classification is not asserted for the coupled law.

3 Decision value in the selected law: a bounded comparison proof

3.1 The auxiliary fixed-work reference

Delete mismatch and retained recovery only for a reference, denoted J_i^0. H1, arrivals, storage, initial information, completion and c_0 are unchanged. A is free because its offer is below R; E is a reflected fluid queue with service $(R - \lambda_A)/c_0$. This is exactly the earlier H1/T1 server, not a replacement proposed for the selected law.

For sufficiently large fixed finite c_0, the earlier complete-time calculation under (5) proves:

1. $J_{\mathcal{L}}^0$ has the unique optimum v_+;
2. a feasible small decrease from v_+ strictly improves $J_{\mathcal{H}}^0$, whereas it strictly worsens $J_{\mathcal{L}}^0$;
3. the two continuous reference objectives have disjoint minimizing sets.

The branch is ensured by $c_0 > 17/3$. More precisely, writing the earlier exact expression as

$$J_i^0 = \frac{c_0 \overline{E}_i(v)^2}{2R} + I_{0,i}(v) + I_{1,i}(v)/c_0, \tag{9}$$

with $\overline{E}_i = hE_{f,i}(1 - e^{-\ell/h})$, $A_+ = \overline{E}_{\mathcal{H}}(v_+)\overline{E}'_{\mathcal{H}}(v_+)/R > 0$, it is sufficient to take

$$c_0 > \max\{17/3, (|I'_{0,\mathcal{H}}(v_+)| + |I'_{1,\mathcal{H}}(v_+)|)/A_+\}. \tag{10}$$

All geometry is fixed first. This recalls the archived exact proof; it does not select an experimental work factor or claim physical support.

3.2 Uniformly bounding the actual mismatch and recovery effects

Now restore the selected W-F/R-E/O-R law with $\beta > 0$ and finite $a_r > 0$. Its isolated loss area at zero mark and a uniform account bound are

$$u_R = U(R), \quad W_0 = \frac{K}{w + u_R} \frac{u_R^2}{2a_r}, \quad 0 \le p(t) \le \varepsilon_p := \frac{RW_0}{mc_0}. \tag{11}$$

Indeed $g_E \le R/c_0$ and the retained-window integral is a subset of the nonnegative time integral, bounded by $(R/c_0)W_0/m$. This argument retains the actual renewal rule and any moving cutoff.

Let $a_* = v_+ \max_i r_i < R$. Take

$$0 < \beta \le 1, \quad c_0 - \beta(1 + v_+/w) > 0, \qquad \varepsilon_p < \min\{(R - a_*)/2, R/2\}. \tag{12}$$

Then $\mathcal{B} \ge R - \varepsilon_p > a_* \ge \lambda_A$: A remains free, not by suppressing its queue, but because its actual offer always fits. The selected E flow while queued lies between

$$\mu_-(t) = \frac{R - \varepsilon_p - \lambda_A(t)}{c_0 + \beta} \le g_E(t) \le \mu_0(t) = \frac{R - \lambda_A(t)}{c_0}. \tag{13}$$

Here $\lambda_A = 0$ outside its pulse. Both comparison services are positive. If E is empty, its free continuation serves its offer or activates only on a compatible strict shortfall. These guards preserve queue comparison.

For the same E arrivals, denote the reflected queues driven by μ_0, μ_- by q_0, q_-. Comparing derivatives at a first crossing of positive queues, and using nonnegative reflection at zero, gives

$$q_0(t) \le q_E(t) \le q_-(t), \qquad q_A(t) = 0. \tag{14}$$

This comparison does not assert that the actual state-dependent queue itself is a fixed-capacity reflection. It accommodates the free/queued speed-mark change at birth.

Moreover,

$$0 \le \mu_0 - \mu_- = \frac{\beta(R - \lambda_A) + c_0\varepsilon_p}{c_0(c_0 + \beta)} \le \varepsilon_\mu := \frac{\varepsilon_p}{c_0} + \frac{R\beta}{c_0^2}. \tag{15}$$

The reflection representation as the supremum over net arrivals on $[s, t]$ implies $0 \le q_- - q_0 \le t\varepsilon_\mu$. Choose a fixed $H > L/v_- + 2(c_0 + 1)M/R$, independent of the small parameters. All compared cohorts clear by H. Integrating the queue bracket proves the new uniform objective estimate

$$0 \le J_i(v) - J_i^0(v) \le \varepsilon_J := \frac{H^2}{2}\varepsilon_\mu \quad \text{for both preparations and every advice.} \tag{16}$$

Since $\varepsilon_p = O(a_r^{-1})$, the bound can be made arbitrarily small with positive β and finite positive a_r. These are mathematical existence choices, not recommended traffic settings.

3.3 A common aggregate action has unavoidable regret

Define the positive reference separation

$$\Gamma_0 = \min_{v \in [v_-, v_+]} \max_{i \in \{\mathcal{L}, \mathcal{H}\}} \{J_i^0(v) - \min_z J_i^0(z)\} > 0. \tag{17}$$

It is positive by continuity, compactness and disjoint minimizing sets. For the actual objectives let $V_i = \inf_v J_i(v)$, making no additional global smoothness or attainment assumption. From (16),

$$\inf_v \max_i \{J_i(v) - V_i\} \ge \Gamma_0 - 2\varepsilon_J > 0 \quad \text{if } \varepsilon_J < \Gamma_0/4. \tag{18}$$

Therefore every deterministic aggregate-only action has nonzero regret for at least one preparation, even allowing optimal lawful inference from the identical full history. A+E and G can distinguish the preparations. The same conclusion concerns arbitrarily near-optimal actions if an infimum is not attained; no general minimum-existence theorem is supplied.

Choose a fixed feasible $\Delta v > 0$ from the strict reference slope contrast. The two reference cost changes between v_+ and $v_+ - \Delta v$ have opposite signs. Taking $2\varepsilon_J$ below both strict gaps preserves those signs for the actual selected law. This establishes opposite *finite-perturbation* usefulness, not automatically opposite infinitesimal gradients at v_+.

Limits of this result. The constructed range may need large work, weak mismatch and fast recovery. A stays free there. The earlier complete A-delay/active-renewal witness is a different range; no joint strong-delay and action-split witness is claimed. Neither range is calibrated or a default evaluation setting. The proof constructs existence transparently; it does not license selecting only favorable evaluation cases.

4 The full sensitivity path on regular event regimes

Use $e_v(z; v)$ for the derivative at fixed physical progress. On each smooth H1 branch its variational equation follows directly from (1); $e_v(0) = 0$ and

$$(e_v)_z = \begin{cases} -e_v/h, & \text{donor}, \\ -(e_v + k_c'(v))/(\gamma h), & \text{receiving}, \end{cases} \qquad k_c'(v) = -K/(w+v)^2. \tag{19}$$

The H1 vector fields match at their switch. Its moving location matters, but there is no state-sensitivity jump from unequal vector fields there.

Each initial position x becomes eligible at $(L-x)/v$. Exactly,

$$P_E(t; v) = \int_0^\ell e(L_p - x; v)\mathbf{1}_{(L-x)/v \le t}\,\mathrm{d}x. \tag{20}$$

For an interior pulse time, put $x_* = L - vt$, $z_* = vt - b$; then

$$\begin{aligned} P_{E,v} &= \int_{x_*}^\ell e_v(L_p - x; v)\,\mathrm{d}x + te(z_*; v), \\ \lambda_E &= ve(z_*; v), \qquad \lambda_{E,v} = e + ve_v + vte_z, \\ \lambda_{A,v} &= r - \lambda_{E,v}, \quad t_f' = -(L-\ell)/v^2, \quad t_r' = -L/v^2. \end{aligned} \tag{21}$$

Before the front $P_{E,v} = 0$; after the rear it is the full integral of e_v. Front/rear changes are treated with their one-sided values. Holding travel delays or arrival times fixed removes a real causal channel.

For queued E, $\mathcal{H}(g; a, u) = g[c_0 + \chi a(u - U(g))_+] = x$. On positive mismatch, its regular implicit derivative is

$$\begin{aligned} &\mathcal{H}_g = c_0 + \chi a[u - U(g) - gU'(g)] \ge \eta > 0, \\ &\mathcal{H}_a = \chi g[u - U(g)], \quad \mathcal{H}_u = \chi a g, \qquad g_v = \frac{x_v - \mathcal{H}_a a_v - \mathcal{H}_u u_v}{\mathcal{H}_g}. \end{aligned} \tag{22}$$

On the inactive side, $\mathcal{H}_g = c_0$, $\mathcal{H}_a = \mathcal{H}_u = 0$. Free A uses $a = \lambda_A$, $u = v$, $x = \mathcal{B} - \lambda_A$. Both queued use $a = x = \mathcal{B}/2$, $u = U(\mathcal{B}/2)$. Thus there is no direct advice term in fully queued local work at fixed budget; advice still changes the budget and arrivals. Differentiating an advice-dependent normalizer would reintroduce the rejected W-A channel.

5 Recovery differentiation must include events and retirement

For fixed physical parameters, write

$$p(t) = \frac{1}{m}\int_{s_m(t)}^{t} D(t-s; u_E(s)) g_E(s)\,\mathrm{d}s, \tag{23}$$

where $s_m = 0$ before fill and, after fill, $Y_E(s_m) = Y_E(t) - m$. Zero-rate intervals carry no admitted mass. On a regular filled-window chart the old cutoff lies at positive E rate and away from a mark jump. Put $Z = \partial_v Y_E$ and $D_o = D(t - s_m; u_E(s_m))$. Differentiating the rank equation gives

$$g_E(s_m)\partial_v s_m = Z(t) - Z(s_m). \tag{24}$$

Let $\tau_k(v)$ be the finite source/mark switching times strictly inside the retained time window, and set $\Phi_k^{\pm}(t) = D(t - \tau_k; u_E(\tau_k^{\pm}))g_E(\tau_k^{\pm})$. Piecewise Leibniz differentiation yields

$$\begin{aligned} p_v(t) = \frac{1}{m}\Bigg[&\int_{s_m}^{t} \{D_u u_{E,v} g_E + D g_{E,v}\}\,\mathrm{d}s + \sum_k (\Phi_k^- - \Phi_k^+)\tau_k' \\ &- \mathbf{1}_{Y_E(t)>m} D_o\{Z(t) - Z(s_m)\}\Bigg], \qquad \mathcal{B}_v = -p_v. \end{aligned} \tag{25}$$

Integral derivatives are within the smooth pieces; the sum accounts for their moving endpoints. The final term is retirement of old rank weight. It has no inverse old-rate factor after cancellation. Omitting it differentiates a different, all-history source. Profile support cutoffs add no jump term because D is continuous and zero there. At a fill, cutoff/mark encounter or rate plateau, use the appropriate event/one-sided analysis; the displayed regular formula is not a universal differentiability theorem at such boundaries.

The active isolated profile uses $S = \sqrt{(w+u)^2 + 2wa_r\theta}$ and has

$$D_u = -Kw(w+u)/S^3, \qquad D_\theta = -Kw^2 a_r/S^3. \tag{26}$$

These are zero on its inactive side. The future update must retain the speed/age history, rather than regard the current meter reading as all dynamic state.

At a transverse continuous-state event with guard $\psi(\tau, X(\tau^-; v), v) = 0$, differentiate the guard and match the same physical state at the shifted event time:

$$\tau' = -\frac{\psi_v + \psi_X X_v^-}{\psi_t + \psi_X F^-}, \qquad X_v^+ = X_v^- + (F^- - F^+)\tau'. \tag{27}$$

For an A birth the guard is its free-feasibility margin $\mathcal{B} - \lambda_A$, not the identically zero free A queue. For a transverse clearance it is the positive queue reaching zero. This is the identity-reset event-timing correction; see Kong et al., Section III-A/B. The history-dependent account still requires our separate (25); a finite-dimensional saltation citation does not prove that history reduction away.

6 Complete TST, regular derivatives and nonregular cases

For each fixed parameter case choose one action-independent horizon

$$H_* > L/v_- + c_{\max} M/B_{\min}, \qquad B_{\min} > 0 \tag{28}$$

from the already proved uniform budget bound. All mass has crossed by it; remaining recovery of already completed traffic need not be integrated. The two equivalent objective and sensitivity

expressions are

$$J = \int_0^{H_*} (M - Y_A - Y_E - Y_B)\,\mathrm{d}t = \frac{M(L-\ell/2)}{v} + \int_0^{H_*} (q_A + q_E)\,\mathrm{d}t,$$
$$J' = -\int_0^{H_*} (Z_A + Z_E + Z_B)\,\mathrm{d}t = -\frac{M(L-\ell/2)}{v^2} + \int_0^{H_*} (q_{A,v} + q_{E,v})\,\mathrm{d}t. \tag{29}$$

Queues are continuous and zero at their episode endpoints, so queue-area endpoint values contribute zero. Their event-time effects on the subsequent state, flows and history do not vanish. In particular, an A birth can change service to E while E already has a positive queue.

Proposition 1 (Regular-chart reference)**.** *On a neighborhood with a fixed finite event sequence, transverse guards, smooth branches and regular renewal cutoffs, the selected local causal construction admits the sensitivities above. They determine the complete-objective derivative, including the terminal clearance tail.*

Proof. Within each history chart, the strictly regular root and piecewise smooth Leibniz map have parameter derivatives. Differentiate the locally uniform Volterra contraction used in the forward construction. This gives the linear causal sensitivity equation. At each of the finitely many transverse events, the implicit-function guard derivative and state matching give (27); split history integrals at those moving times. Induct through the fixed event list. Finite bounds on the chart justify differentiating the fixed-horizon objective. This proves a local statement, not regularity of every advice or event sequence. □

Forward-admitted grazing, coincident events, queue-activation thresholds, renewal fill/cutoff encounters and root/profile kinks remain in reporting. Check continuity and feasible one-sided difference behavior separately. If no finite ordinary or one-sided derivative exists, label it so and use actual finite-perturbation costs for that comparison. Do not smooth, freeze events, delete boundary cases or reinterpret a jump quotient as a correct gradient. No general continuity, global minimum attainment or opposite infinitesimal slopes for the selected coupled objectives is proved by the uniform value result in Section 3.

The literature basis is deliberately bounded. Kong et al. explains the event-timing correction and its transversality limits; it is not a traffic or infinite-history theorem. Delle Monache et al. establishes that VSL cost variations and gradient control already have a literature precedent, under a different LWR/outflow-tracking task; its equations are not our TST gradient. The model-specific arrival, root, renewal and comparison derivations here are our deductions from the accepted law.

7 Verification contract and present evidence limit

Later execution must distinguish the continuous analytical derivative, the implemented forward model's directional differences and its AD derivative. Agreement of the last two alone does not prove convergence to the defined model. Use double precision, feasible step/refinement ladders, event signatures, complete clearance and both forms of (29). Validate the value/change of actual complete TST against the predicted direction and a strong scalar-search reference; information advantage does not imply AD or lane-state superiority.

This review checks symbolic identities and exact static inequalities only. No numerical traffic/ODE trajectory, finite-difference gradient, AD, calibration or policy optimization is performed. Numerical tolerances and resource/solver settings belong to a separately approved execution plan. The positive existence and negative interval are both retained; neither chooses the main evaluation parameters or establishes physical prevalence.

References

[1] Archived own derivation. Step 3: information and full-time sensitivity, 14 October 2026, Sections 1–7. Used only as the explicitly deleted-mechanism reference and predecision loading construction.

[2] Archived own derivation. Full current mathematics, 16 October 2026. Selected equations and conditional forward bounds.

[3] N. J. Kong, J. J. Payne, J. Zhu and A. M. Johnson. *Saltation Matrices: The Essential Tool for Linearizing Hybrid Dynamical Systems*. Proceedings of the IEEE 112(6), 585–608, 2024. Author manuscript, v3, Section III-A/B; DOI 10.1109/JPROC.2024.3440211.

[4] M. L. Delle Monache, B. Piccoli and F. Rossi. *Traffic regulation via controlled speed limit*. arXiv:1603.04785, 2016, abstract and Section 3.3. Different control/objective; precedent only.